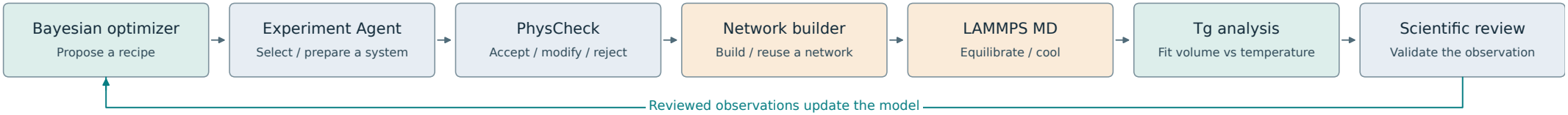
Arthur da Silva Sousa Santos, Adib Bazgir, Elena Stojanovska, Nannan Zhang, Xiaoli Yan, Matheus Rodrigues de Amorim Medina

Code: github.com/elensto/Autonomous_Formulation_Optimizer

Closing the loop from epoxy formulation to glass transition



Goal. Find DGEBA/4,4′-DDS epoxy formulations whose simulated T_g falls in a 150–220 °C target window. Each loop step spends HPC time only on candidates that are chemically valid and that the model rates as the most informative.

1 | Bayesian optimizer. The design variables are the amine-H/epoxy ratio $r = 4N_{\mathrm{DDS}}/2N_{\mathrm{DGEBA}}$ and the epoxy conversion. With no data, the loop starts from the formulation closest to the 1:1 recipe. Farthest-first exploration follows until there are four distinct accepted points. After that, a Matérn-5/2 Gaussian process, weighted by the uncertainty of each observation, ranks candidates by $P(T_g \in [150, 220]\,^{\circ}\mathrm{C})$. Completed and running candidates are never proposed again.

2 | Experiment Agent and PhysCheck. The agent turns each proposal into a simulation request and returns *accept*, *modify* or *reject*. Deterministic gates check: consistent stoichiometry; conversion $\leq 100\min(1, r)$%; an existing buildable network recipe; matching epoxy and amine-H counts; charge neutrality ($|q| < 10^{-3}\,e$); system size; structure hash; and force-field/protocol compatibility.

3 | Network preparation and MD. We built 23 formulations (r = 0.80–1.14) by removing whole molecules from a published 100 DGEBA + 50 DDS box (6,350 atoms) [1]. Coordinates, charges and parameters of the remaining atoms were left unchanged. Curing used LAMMPS `fix bond/react` at 503 K and 1 atm up to 90% of the limiting reagent, with annealing between reactive cycles. The screening protocol was 2 ns NPT at 503 K (density), a 0.5 ns hold at 800 K, then 26 plateaus down to 300 K, each with 50 ps equilibration and 50 ps sampling (≈200 K ns^{-1}). We benchmarked a pretrained MLIP (PET-MAD) at 0.6–1.8 s per step, which would take more than 100 days per T_g scan. We therefore ran production with the classical force field.

4 | T_g analysis and post-PhysCheck. We fit two lines to specific volume versus T and take T_g as their intersection. 200 block-bootstrap resamples give the sampling interval. A result is accepted only if T_g lies inside its own interval, the 95% interval is narrower than 80 °C, the achieved conversion is within 5 points of the request and does not exceed the stoichiometric limit, and T_g is physically plausible.

5 | Observation store and learning. Accepted results are written to a store that the optimizer can read directly. They are tagged trainable or non-trainable and logged with the job request, raw result and every gate decision. The next iteration retrains on them. A `replay` backend reproduces the loop on the completed MD runs; a `slurm` backend writes cluster jobs for new candidates.

Results. The eight completed networks give preliminary T_g values from 196.1 to 296.4 °C at 73.0–90.4% epoxy conversion; density at 503 K is 1.07–1.10 g cm^{-3}. All eight curves show two clear linear branches and passed the automated post-PhysCheck. **100 DGEBA + 45 DDS (r = 0.90, 82% conversion) reaches T_g = 196.1 °C (189.9–201.9 °C), inside the target window.** In the closed-loop replay, the optimizer selected this formulation on its **3rd** simulation, compared with 4.5 expected for random selection. Once the GP took over, new results tested its predictions: it gave D100_H044 $P = 0.68$ of landing in the window, and the measured 240.5 °C updated the model.

Limitations and next steps. Fast MD cooling (∼$10^{10}\times$ experimental rates) inflates absolute T_g. Each formulation is a single network realization, and formal scientific review is pending before the GP is used in production. Next: replicas of the hit, live `slurm` runs of the 15 remaining formulations, and MLIP validation of the force field.

[1] Livraghi et al., *Block Chemistry for Accurate Modeling of Epoxy Resins*, J. Phys. Chem. B 127 (2023), 7648–7662, doi:10.1021/acs.jpcb.3c04724; data: Zenodo 7273800 (CC BY 4.0).

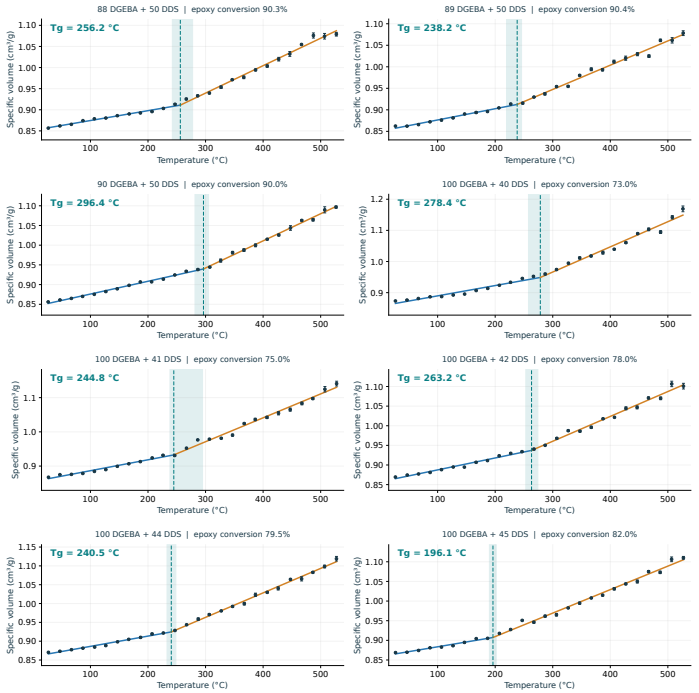


Figure 1. Simulated specific volume versus temperature for eight DGEBA/DDS formulations. Points are plateau means with block standard errors. The fitted glassy (blue) and rubbery (orange) branches intersect at T_g (dashed line); shading shows its 95% bootstrap interval, which covers sampling only. Panel titles give molecule counts and epoxy conversion.

Formulation	DGEBA : DDS	r	Conv. (%)	T_g (°C)	95% interval (°C)	$\rho_{503\,\mathrm{K}}$ (g cm^{-3})	Loop order
D088_H050	88 : 50	1.136	90.3	256.2	242.3–277.5	1.100	7
D089_H050	89 : 50	1.124	90.4	238.2	219.7–245.8	1.090	8
D090_H050	90 : 50	1.111	90.0	296.4	281.7–305.2	1.094	1
D100_H040	100 : 40	0.800	73.0	278.4	257.9–294.0	1.072	2
D100_H041	100 : 41	0.820	75.0	244.8	238.2–294.8	1.080	6
D100_H042	100 : 42	0.840	78.0	263.2	253.2–274.1	1.082	4
D100_H044	100 : 44	0.880	79.5	240.5	233.1–248.4	1.084	5
D100_H045	**100 : 45**	**0.900**	**82.0**	**196.1**	**189.9–201.9**	**1.092**	**3**

Table 1. Preliminary MD results. The intervals reflect sampling within each trajectory only. “Loop order” is the order in which the closed loop selected each formulation in the replay run. The shaded row is the only formulation inside the target window.